

Co-Amoxiclav 1.2 g Injection
Amoxicillin 1000 mg
Clavulanic Acid 200 mg

Product Name

Co-Amoxiclav for Injection 1.2 g

Composition

Each vial contains:

Amoxicillin sodium equivalent to Amoxicillin 1000 mg

Potassium clavulanate equivalent to Clavulanic Acid 200 mg

Product Description

Almost white-colored powder for injection.

A pale yellow color solution after reconstitution.

Pharmacodynamics

Amoxicillin is an orally active member of the penicillin family. In amoxicillin, the benzyl ring in site change extends the range of antimicrobial activity into the gram-negative bacteria. It is bactericidal for both gram-positive and negative bacteria.

Clavulanate is a β -lactam antibiotic and interacts with β -lactamase, this interaction may lead to enzyme induction, inactivation of the enzyme, and hydrolysis of the β -lactam ring is used clinically only in combination with amoxicillin.

Amoxicillin kills bacteria by interfering with the synthesis of the bacteria cell wall. As the lactamase enzyme secreted by the bacteria destroys amoxicillin the drug is ineffective in most staphylococcal infections. An increasing percentage of *Salmonella* spp., *E. coli*, *Proteus mirabilis*, *Shigella* spp. are not sensitive to amoxicillin. The spectrum of activity of amoxicillin may be extended by the concomitant use of β -lactamase inhibitors such as clavulanic acid. Clavulanic acid has been reported to enhance the activity of amoxicillin, against several species not generally considered sensitive. Clavulanate by itself has little antibacterial activity. However, in association with amoxicillin, it produces an antibiotic agent of broad-spectrum.

Co-Amoxiclav is bactericidal to a wide range of organisms such as:

Gram-positive:

Aerobes: **Enterococcus faecalis*, *Streptococcus pneumoniae*, *Streptococcus pyogenes*, *Streptococcus viridans*, **Staphylococcus aureus*, **Coagulase-negative staphylococci* (including *Staphylococcus epidermidis*), *Corynebacterium* sp, **Bacillus anthracis*, *Listeria monocytogenes*, *Enterococcus faecium*, *Streptococcus agalactiae*, *Streptococcus* sp.

Anaerobes: *Clostridium*, *Peptococcus* spp, *Peptostreptococcus*.

Gram-negative:

Aerobes: **Haemophilus influenzae*, **Escherichia coli*, **Proteus mirabilis*, **Proteus vulgaris*, **Klebsiella* sp, **Moraxella catarrhalis* (*Branhamella catharralis*), **Salmonella* sp, **Shigella* sp, *Bordetella pertussis*, *Brucella* sp, **Neisseria gonorrhoeae*, **Neisseria meningitidis*, *Vibrio cholerae*, **Yersinia enterocolitica*, *Pasteurella multocida*, *Gardnerella vaginalis*, *Helicobacter pylori*, *Legionella* sp.

Anaerobes: **Bacteroides* sp (including *Bacteroides fragilis*), **Fusobacterium* sp.

*Including β -lactamase producing strains resistant to ampicillin and amoxicillin.

Infections caused by amoxicillin-susceptible organisms are amenable to Co-Amoxiclav treatment due to their amoxicillin content. Mixed infections caused by amoxicillin- susceptible organisms in conjunction with Co-Amoxiclav-susceptible β -lactamase- producing organisms may therefore be treated with Co-Amoxiclav.

Pharmacokinetics

Amoxicillin is rapidly distributed throughout the body. The volume of distribution is 0.3 lit/kg body weight. Plasma protein binding is around 20%.

Amoxicillin is metabolized to a limited extent to penicilloic acid, which is excreted in the urine. After parenteral administration, 75% is excreted via the kidney in the following 6 hours.

Indication

Co-Amoxiclav is for short-term treatment of bacterial infections at the following sites:

- Upper respiratory tract infections [including eyes, ears, nose (EENT)] e.g., recurrent tonsillitis, sinusitis, otitis media.
- Lower respiratory tract infections e.g., acute exacerbation of chronic bronchitis, lobar, and bronchopneumonia.
- Skin and soft tissue infections e.g., boils, abscesses, cellulitis, wound infections. Bone and joint infections e.g., osteomyelitis.
- Other infections e.g., dental infections, septic abortion, puerperal sepsis, intra-abdominal sepsis. Also septicemia, peritonitis, and postsurgical infections.

Co-Amoxiclav is also indicated for prophylaxis against infection which may be associated with major surgical procedures e.g., gastrointestinal, pelvic, head and neck, cardiac, renal, joint replacement, and biliary tract.

Recommended Dosage

Dosage & treatment for Infections

Adults and Children >12 years:

- Usually 1.2 g 8 hourly. In more serious infections, increase frequency to 6-hourly intervals.

Children 3 months to 12 years:

- Usually 30 mg/kg* 8 hourly. In more serious infections, increase frequency to 6-hourly intervals.

Children 0-3 months:

- 30 mg/kg* every 12 hrs in premature infants and full-term infants during the perinatal period, increasing to 8 hrs thereafter.

*Each 30-mg Co-Amoxiclav contains amoxicillin 25 mg and clavulanic acid 5 mg.

Dosage for Surgical Prophylaxis:

Adults:

The usual dose is Co-Amoxiclav 1.2g at the induction of anesthesia. Operations where there is a high risk of infection e.g., colorectal surgery, may require 3, and up to 4 doses of Co-Amoxiclav 1.2g in a 24-hr period. These doses are usually given at 0, 8, 16 (and 24) hrs. This regimen can be continued for several days if the procedure has a significantly increased risk of infection.

Clear clinical signs of infection at operation will require a normal course of IV or oral Co-Amoxiclav therapy postoperatively.

Renal Impairment

Adults: Mild Impairment (creatinine clearance >30 mL/min):

No change in dosage.

Moderate Impairment (creatinine clearance 10-30 mL/min):

- 1.2 g IV stat, followed by 600 mg IV 12 hourly.

Severe Impairment (creatinine clearance <10 mL/min):

Adult:

- 1.2 g IV stat, followed by 600 mg IV 24 hourly. Dialysis decreases serum concentrations of Co-Amoxiclav and an additional 600-mg IV dose may need to be given during dialysis and at the end of dialysis.

Children:

Similar reductions in dosage should be made.

Hepatic Impairment:

Dose with caution; monitor hepatic function at regular intervals.

Each 1.2-g vial Co-Amoxiclav contains 1 mmol of potassium and 3.1 mmol of sodium (approximately).

Route of administration

Co-Amoxiclav for Injection BP 1.2 g may be administered either by slow intravenous injection over a period of 3 to 4 minutes directly into a vein or via a drip tube or by infusion over 30 to 40 minutes. It is not suitable for intramuscular administration. Children aged less than 3 months should be administered by infusion only.

Method of administration

IV Injection: 1 vial of Co-Amoxiclav for Injection 1.2 g diluted with 20 mL of Water for Injection and administered by slow intravenous injection over a period of 3 to 4 min directly into a vein via a drip tube. A transient pink coloration may appear during reconstitution. Reconstituted solutions are normally a pale yellow colour and should be discarded if not used within 20 minutes.

IV Infusion: 1 reconstituted vial of Co-Amoxiclav for Injection 1.2 g diluted to a minimum volume of 100 mL of infusion fluid (Water for Injection, 0.9 % Sodium Chloride Infusion, Compound Sodium Chloride Injection 1959 [Ringer's], Compound Sodium Lactate Intravenous infusion [Ringer-Lactate: Hartmann's] and 0.3 % w/v Potassium Chloride & 0.9 % w/v Sodium Chloride Intravenous infusion) by infusion over 30 to 40 min.

IV Injection: The stability of the solution is concentration-dependent, thus it should be used immediately upon reconstitution and given by slow IV injection over a period of 3-4 min. Co-Amoxiclav solutions should be used within 20 min of reconstitution. It may be injected directly into a vein or via a drip tube.

IV Infusion: Alternatively, Co-Amoxiclav may be infused in the infusion fluids mentioned in Route of Administration (e.g., using a mini-bag or in-line burette) without delay⁺ by infusion over 30 to 40 minutes.

⁺Solutions should be made up to full infusion volume immediately after reconstitution.

Any residual antibiotic solutions should be discarded.

Contraindications

Co-Amoxiclav is contraindicated in patients with a history of hypersensitivity to penicillin and in patients who are sensitive to amoxicillin & potassium injection-associated cholestatic jaundice or hepatic dysfunction. Patients with suspected glandular fever, pharyngitis should not be given amoxicillin & potassium injection as the amoxicillin component is likely to cause a maculopapular rash.

Warnings and Precautions

Before initiating the amoxicillin & potassium therapy, a thorough examination of the patient's health record has to be made to ensure that the patient is not allergic to penicillin, cephalosporin, or other allergens. In case of the

occurrence of allergy with amoxicillin & potassium, the medication should be stopped and the appropriate therapy to be initiated. Pseudomembranous colitis has been reported with antibacterial treatments including amoxicillin & potassium clavulanate. In moderate to severe cases, consideration should be given to management with fluids and electrolytes and treatment with an antibacterial drug clinically effective against *Clostridium difficile* associated colitis. Avoid alcohol (includes wine, beer, and liquid) as alcohol can interfere in the treatment of injection. Diabetic patients should be cautious as glucose urine tests may give false-positive results while using amoxicillin & potassium.

Use in patients with impaired function:

The use of amoxicillin & potassium has been associated with cholestatic jaundice and changes in liver function tests.

Dose with caution, monitoring the liver function at regular intervals.

Use in elderly patients:

No dose adjustments are required.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Co-Amoxiclav, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems, or other beta-lactam agents. If an allergic reaction occurs, Co-Amoxiclav must be discontinued immediately and appropriate alternative therapy instituted.

Interactions with Other Medicaments

Concomitant use of probenecid is not recommended. Probenecid decreases the renal tubular secretion of amoxicillin. Concomitant use with Co-Amoxiclav may result in increased and prolonged blood levels of amoxicillin but not clavulanate.

Concomitant use of allopurinol during treatment with amoxicillin can increase the likelihood of allergic skin reactions. There are no data on the concomitant use of Co-Amoxiclav and allopurinol.

In common with other antibiotics, Co-Amoxiclav may affect the gut flora, leading to lower estrogen reabsorption and reduced efficacy of combined oral contraceptives.

Pregnancy and Lactation

Reproduction studies in animals (mice and rats) with orally and parenterally administered Co-Amoxiclav have shown no teratogenic effects. There is limited experience of the use of Co-Amoxiclav in human pregnancy. As with all medicines, use should be avoided in pregnancy, especially during the 1st trimester, unless considered essential by the physician.

Co-Amoxiclav may be administered during the period of lactation. Except for the risk of sensitization associated with the excretion of trace quantities in breast milk, there are no detrimental effects for the breastfed infant.

Side Effects / Adverse Effects

The following convention has been used for the classification of frequency:

Very common, common, uncommon, very rare.

Infections and Infestations:

Common: Mucocutaneous candidiasis.

Blood and Lymphatic System Disorders:

Rare: Reversible leucopenia (including neutropenia) and thrombocytopenia.

Very Rare: Reversible agranulocytosis, hemolytic anaemia, prolongation of bleeding time, and prothrombin time.

Immune System Disorders:

Very Rare: Angioneurotic edema, anaphylaxis, serum sickness-like syndrome, hypersensitivity vasculitis.

Very Rare: Reversible hyperactivity and convulsions may occur in patients with impaired renal function or those receiving high doses.

Gastrointestinal Disorders:

Adults:

Very Common: Diarrhoea.

Common: Nausea, vomiting.

Very rare: Antibiotic-associated colitis (including pseudomembranous colitis and hemorrhagic colitis) is less likely to occur after parenteral administration

Children:

Common: Diarrhoea, nausea, vomiting.

All Populations:

Nausea is more often associated with higher oral dosages. If gastrointestinal reactions are evident, they may be reduced by taking Co-Amoxiclav at the start of a meal.

Uncommon: Indigestion.

Very Rare: Antibiotic-associated colitis (including pseudomembranous colitis and hemorrhagic colitis). Black hairy tongue.

Hepatobiliary Disorders:

Uncommon: A moderate rise in AST and/or ALT has been noted in patients treated with β -lactam class antibiotics, but the significance of these findings is unknown.

Very Rare: Hepatitis and cholestatic jaundice.

These events have been noted with other penicillins and cephalosporins. Hepatic events have been reported predominantly in males and elderly patients and may be associated with prolonged treatment. These events have been rarely reported in children.

Signs and symptoms usually occur during or shortly after treatment but in some cases may not become apparent until several weeks after treatment has ceased. These are usually reversible. Hepatic events may be severe and in extremely rare circumstances, deaths have been reported. These have almost always occurred in patients with serious underlying disease or taking concomitant medications known to have the potential for hepatic effects.

Skin and Subcutaneous Tissue Disorders:

Uncommon: Skin rash, pruritus, urticaria.

Rare: Erythema multiforme.

Very Rare: Stevens-Johnson syndrome, toxic epidermal necrolysis, bullous exfoliative dermatitis, acute generalized exanthematous pustulosis (AGEP).

Very rare: Drug reaction with Eosinophilia and Systemic Symptoms (DRESS).

If any hypersensitivity dermatitis reaction occurs, treatment should be discontinued.

Renal and Urinary Disorders:

Very Rare: Intestinal nephritis, crystalluria.

IV: Nervous System Disorders:

Uncommon: Dizziness, headache.

Very Rare: Reversible hyperacidity and convulsions may occur in patients with impaired renal function or those receiving high doses.

Symptoms and Treatment of Overdose

Gastrointestinal symptoms and disturbance of the fluid and electrolyte balances may be evident. Gastrointestinal symptoms may be treated symptomatically with attention to the water-electrolyte balance.

Amoxicillin crystalluria, in some cases leading to renal failure, has been observed.

Co-Amoxiclav can be removed from the circulation by hemodialysis.

Amoxicillin has been reported to precipitate in bladder catheters after IV administration of large doses. A regular check of potency should be maintained.

Incompatibilities

Co-Amoxiclav should not be mixed with amino acid solutions, lipid emulsions, blood, or glucose. It is unstable in dextran or bicarbonate infusions. Must be injected into drip tubing.

Co-Amoxiclav should not be mixed with blood products, other proteinaceous fluids i.e., protein hydrolysates or with IV lipid emulsions.

If Co-Amoxiclav is prescribed concurrently with an aminoglycoside, the antibiotics should not be mixed in the syringe, IV fluid container or given set because loss of activity of the aminoglycoside can occur under these conditions.

Storage Condition

Store at temperature below 30°C. Protect from light. Keep out of reach of children.

Stability of the Reconstituted Solution

For 20 minutes with water for injection (concentrated solution for intravenous injection) at room temperature.

For 40 minutes with water for injection, 0.9% w/v sodium chloride infusion, compound sodium chloride injection 1959 (Ringer's), compound sodium lactate intravenous infusion (Ringer-lactate: Hartmann's) and 0.3% w/v potassium- chloride and 0.9% w/v sodium chloride intravenous infusion (diluted solutions for intravenous infusion) at room temperature.

Shelf Life

24 months from the date of manufacture if kept as recommended.

Presentation

1 or 10 vials are packed in a folded paper box.

Name and Address of Manufacturer

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